Problem 1: The Quantum Node Hunter

- **Analytical Task:** Mathematically prove that the wavefunction for energy state n has exactly $n - 1$ nodes (points inside the well where the probability of finding the particle is exactly zero, excluding the boundaries).
- **Coding Task:** Modify the script to simulate $n = 5$. Use your generated graph to visually confirm the number and exact x-coordinates of the internal nodes.

Problem 2: Finding the Particle

- **Analytical Task:** For a particle in its first excited state ($n = 2$), calculate the exact theoretical probability of finding the particle in the left quarter of the box (between $x = 0$ and $x = L/4$). *Hint: Integrate $|\psi_2(x)|^2$ from 0 to $L/4$.*
- **Coding Task:** Use numerical integration in your script (e.g., using `numpy.trapz` or `scipy.integrate.simpson`) to calculate this area under the $n = 2$ curve and check if your code matches your math.